Polygon Interior and Exterior Angle Sums

Answer Key

High School Geometry

Quick Answers

1. 720
2. 1620
3. 40
4. 98
5. 15
6. 93
7. (a) each exterior angle = 24, (b) number of sides = 15
8. 12
9. (a) correct interior angle sum = 540, —
10. (a) each exterior angle = 30, (b) number of sides = 12

What is verified

12 of 13 answers machine-checked against SymPy, across 10 problems.
— marks an answer only you can judge — problem 9. The worked solution states what a correct response must contain.

Curriculum

Level: High School Geometry
HSG-CO.C – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 13 tagged checks.

Common wrong answers

9. *If they got 900: multiplied the number of sides by 180 without subtracting 2 first*

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1. Find the sum of the measures of the interior angles of a hexagon (6 sides).

Solution: A polygon with n sides splits into $n-2$ triangles from one vertex, so the interior angles add to $(n-2)\cdot 180°$. With $n=6$: $(6-2)\cdot 180° = 4\cdot 180°$. $\boxed{720°}$

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2. Find the sum of the measures of the interior angles of a polygon with 11 sides.

Solution: Same rule with $n=11$: $(11-2)\cdot 180° = 9\cdot 180°$. $\boxed{1620°}$

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3. A regular polygon has 9 sides. Find the measure of each exterior angle.

Solution: One exterior angle at each vertex, taken in the same rotational direction, adds to one full turn: the exterior angles of any convex polygon sum to 360°. In a regular polygon they are all equal, so each is 360°/9. $\boxed{40°}$

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4. Three of the angles of a quadrilateral measure 112°, 87° and 63°. Find the fourth.
Solution: A quadrilateral has $n = 4$, so its angles add to $(4-2)\cdot 180° = 360°$.

$$\begin{aligned} 112 + 87 + 63 + x &= 360 \\ 262 + x &= 360 \\ x &= 98 \end{aligned}$$

The fourth angle is $\boxed{98°}$

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5. Each exterior angle of a regular polygon measures 24°. How many sides does it have?
Solution: The n equal exterior angles fill 360°, so $360/n = 24$, giving $n = 360/24$. $\boxed{n = 15}$

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6. Four of the angles of a pentagon measure 102°, 118°, 96° and 131°. Find the fifth.
Solution: A pentagon has $n = 5$, so its angles add to $(5-2)\cdot 180° = 540°$.

$$\begin{aligned} 102 + 118 + 96 + 131 + x &= 540 \\ 447 + x &= 540 \\ x &= 93 \end{aligned}$$

The fifth angle is $\boxed{93°}$

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7. Each interior angle of a regular polygon measures 156°. (a) Each exterior angle? (b) How many sides?
Solution (a): An interior angle and the exterior angle at the same vertex form a straight line, so they are supplementary: $180° - 156°$. $\boxed{24°}$
Solution (b): The equal exterior angles fill 360°, so $n = 360°/24°$. $\boxed{n = 15}$

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8. The interior angles of a polygon add up to 1800°. How many sides does the polygon have?
Solution: Read the interior-angle-sum rule backwards.

$$\begin{aligned} (n-2)\cdot 180 &= 1800 \\ n - 2 &= 10 \\ n &= 12 \end{aligned}$$

The polygon has $\boxed{n = 12}$

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9. Marcus writes sum $= 5 \cdot 180° = 900°$ for a pentagon. (a) Correct sum? (b) Explain his error.

Solution (a): A pentagon has $n = 5$, and the rule subtracts two before multiplying: $(5 - 2) \cdot 180° = 3 \cdot 180°$. $\boxed{540°}$

Solution (b) — instructor-judged. A model response: drawing the diagonals from one vertex of a pentagon cuts it into 3 triangles, not 5, because the two sides that meet at the chosen vertex do not produce triangles of their own. Marcus multiplied the number of sides by 180°; he should have multiplied *two fewer than* the number of sides, that is $(5-2) \cdot 180°$.

Full credit: names the correct set-up $(5 - 2) \cdot 180°$ and says why the count drops by two (three triangles, not five). *Half credit:* says 900° is too large without identifying which factor is wrong. *No credit:* restates the correct sum with no account of the error.

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10. In a regular polygon each interior angle is 5 times each exterior angle. (a) Each exterior angle? (b) Sides?

Solution (a): Let the exterior angle be x. The interior angle is then $5x$, and the two are supplementary.

$$\begin{aligned} 5x + x &= 180 \\ 6x &= 180 \\ x &= 30 \end{aligned}$$

Each exterior angle measures $\boxed{30°}$

Solution (b): The equal exterior angles fill 360°, so $n = 360°/30°$. $\boxed{n = 12}$